

Stalnaker's static triviality result

Cian Dorr

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1. The concept of probabilistic independence

Definition: When $\Pr$ is a probability function and A and B are propositions in its domain, A and B are *independent in* $\Pr$ iff $\Pr(A \wedge B) = \Pr(A)\Pr(B)$.

Some facts about probabilistic independence:

Fact 1: if either $\Pr(A)$ or $\Pr(B)$ is 0 or 1, A and B are automatically independent in $\Pr$.

Fact 2: The following are all equivalent:

- a) A and B are independent in $\Pr$
 - b) A and $\neg B$ are independent in $\Pr$
 - c) $\neg A$ and B are independent in $\Pr$
 - d) $\neg A$ and $\neg B$ are independent in $\Pr$
- Proof for (b): $\Pr(A \wedge \neg B) = \Pr(A) - \Pr(A \wedge B) = \Pr(A) - \Pr(A)\Pr(B) = \Pr(A)(1 - \Pr(B)) = \Pr(A)\Pr(\neg B)$. Proofs for others are similar.

Fact 3: When $\Pr(B) > 0$, A and B are independent in $\Pr$ iff $\Pr(A) = \Pr(A | B)$.

- So, if $\Pr(Y | X) = \Pr(Y)$ and $\Pr(X) < 1$ then $\Pr(Y | \neg X) = \Pr(Y)$.

Fact 4: if X and Y are independent in $\Pr$ and X logically entails Y , then either $\Pr(X) = 0$ or $\Pr(Y) = \Pr(Y | X) = 1$.

2. Equationality and independence

Where $\Rightarrow$ is any binary operation on propositions:

$\Rightarrow$ obeys MP iff for any propositions A and B , $A \Rightarrow B$ entails $A \supset B$.

$\Rightarrow$ obeys And-to-if iff for any propositions A and B , $A \wedge B$ entails $A \Rightarrow B$.

Fact 5: If $\Rightarrow$ obeys MP and And-to-if, then for any A and B , $A \wedge (A \Rightarrow B)$ is logically equivalent to $A \wedge B$.

Fact 6: If $\Rightarrow$ obeys MP and And-to-if, then for any A and B , $\Pr(A \Rightarrow B | A) = \Pr(B | A)$.

Fact 7: If $\Rightarrow$ obeys MP and And-to-if and is equational for $\Pr$ then for any A and B with $\Pr(A) > 0$, $\Pr(A \Rightarrow B | A) = \Pr(A \Rightarrow B)$ —in other words, $A \Rightarrow B$ is independent of A in $\Pr$.

- (Recall from the last handout the following definition: $\Rightarrow$ is equational for $\Pr$ iff for any A and B in the domain of $\Pr$ with $\Pr(A) > 0$, $\Pr(A \Rightarrow B) = \Pr(B | A)$.)

Fact 8: If $\Rightarrow$ obeys MP and And-to-if and is equational for $\Pr$, then for any A and B such that $A \Rightarrow B$ entails A , either $\Pr(A \Rightarrow B) = 0$ or $\Pr(A) = 1$. (From facts 4 and 7.)

- Strategy: we're going to find schemas A^* and B^* that we can substitute for A and B such that every instance of $A^* \Rightarrow B^*$ logically entails A^* , but so long as Pr is nontrivial there are values of A^* and B^* such that neither $\text{Pr}(B^* | A^*) = 0$ nor $\text{Pr}(A^*) = 1$.

3. "Very Limited Antecedent Strengthening"

Recall the following principle (valid in Lewis's conditional logic, and hence in all the other logics we have looked at):

$$\text{RCV: } A > B, A > C \vdash (A \wedge B) > C$$

or equivalently (given CC and weakening of the consequent):

$$A > B \wedge C \vdash A \wedge B > C$$

or equivalently (given ID):

$$A > (A \wedge B \wedge C) \vdash (A \wedge B) > (A \wedge B \wedge C)$$

or equivalently as a metarule:

$$\text{If Strong} \vdash \text{Medium and Medium} \vdash \text{Weak, then Weak} > \text{Strong} \vdash \text{Medium} > \text{Strong.}$$

4. A conditional that logically entails its own antecedent

Here's our " $A^* \Rightarrow B^*$ ":

$$(S) \quad (A \vee A \Rightarrow B) \Rightarrow A \wedge B$$

by RCV, (S) entails $(A \wedge (A \vee A \Rightarrow B)) \Rightarrow A \wedge B$

by substitution in the antecedent, this entails $A \Rightarrow A \wedge B$

by weakening of the consequent, this entails $A \Rightarrow B$,

by disjunction introduction, this entails $A \vee (A \Rightarrow B)$. QED!

5. Upshot

If $\Rightarrow$ is equational in Pr and obeys minimal conditional logic, then for any A and B , either

$$(a) \text{Pr}(A \vee A \Rightarrow B) = 1, \text{ or}$$

$$(b) \text{Pr}((A \vee A \Rightarrow B) \Rightarrow (A \wedge B)) = 0.$$

If (a), then either (a.1) $\text{Pr}(A) = 1$, or (a.2) $\text{Pr}(A \Rightarrow B | \neg A) = 1$

If (a.2) then $\text{Pr}(A \Rightarrow B) = 1$, so $\text{Pr}(B | A) = 1$, so $\text{Pr}(A \wedge B) = \text{Pr}(A)$.

If (b), then either (b1) $\text{Pr}(A \vee A \Rightarrow B) = 0$ or (b2) $\text{Pr}(A \wedge B | A \vee A \Rightarrow B) = 0$

If (b1) then $\text{Pr}(A) = 0$

If (b2) then $\text{Pr}(A \wedge B \wedge (A \vee A \Rightarrow B)) = 0$, so $\text{Pr}(A \wedge B) = 0$.

So: for any A and B, either $\Pr(A) = 0$, or $\Pr(A) = 1$, or $\Pr(A \wedge B) = 0$, or $\Pr(A \wedge B) = \Pr(A)$

So: there are no A and B such that $0 < \Pr(A \wedge B) < \Pr(A) < 1$.

- That is: Pr is “trivial”.

6. Some possible reactions to this result

(i) Give up altogether on the Equation

(ii) Give up on the idea that conditionals express propositions (up next)

(iii) Weaken the logic by rejecting RCV

- This avenue is explored in Andrew Bacon, ‘Stalnaker’s Thesis in Context’

(iv) Restrict the Equation

- The important inspiration here is from van Fraassen, ‘Probabilities and Conditionals’.

7. More on van Fraassen’s result

Required concepts: some propositions are *categorical* (‘unconditional’, ‘just about how things actually are’...); the rest we can call *hypothetical*.

- When A and B are categorical, $\neg A$, $A \wedge B$, and $A \vee B$ are categorical too, but the propositions eligible to be expressed by ‘If A, B’ are not (typically).
- Simple ordinary sentences not involving ‘if’ express categorical propositions.

Van Fraassen shows that the following are consistent:

Restricted Contextualist Chance Equation: In a context where t is salient, the counterfactual ‘if’ is naturally interpreted as expressing a binary operation $\Rightarrow$ such that for all categorical A having positive chance at t and for all B, the chance at t of $A \Rightarrow B$ equals the conditional chance at t of B given A.

Restricted Contextualist Credence Equation: In a context where person S and time t are salient (e.g. when S is the speaker and t is the time of speech), the indicative ‘if’ is naturally interpreted as expressing a binary operation $\Rightarrow$ such that for all categorical A to which S assigns positive credence at t and for all B, the S’s credence at t in $A \Rightarrow B$ equals S’s conditional credence at t in B given A, provided S is rational.